

MAT 523: Functions of a Real Variable I

Homework V

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(1.)

Statement

Every interval (open, closed, half-open, bounded, unbounded, etc.) is a Borel set.

Proof

The collection $\mathcal{B}$ of Borel sets is the smallest σ -algebra of sets of real numbers that contain all the open sets of real numbers. Since open intervals are open sets, they are Borel sets. Let a be a finite real number. From our previous conclusion, we know that $(a, \infty) \in \mathcal{B}$ (since it is an open interval). But $\mathcal{B}$ is a σ -algebra, so it is closed with respect to the formation of complements. Thus, $(-\infty, a] \in \mathcal{B}$. By similar reasoning, we can conclude that intervals of the form $[a, \infty)$ are members of $\mathcal{B}$. Next, since $\mathcal{B}$ is closed under countable unions, sets of the form $(-\infty, a) \cup (a, \infty)$ are members of $\mathcal{B}$. But then complements of these types of sets are also in $\mathcal{B}$, so

$$\begin{aligned} ((-\infty, a) \cup (a, \infty))^C &= [a, \infty) \cap (-\infty, a] \\ &= \{a\} \end{aligned}$$

is in $\mathcal{B}$. Using this result, we can show that closed intervals are members of $\mathcal{B}$. Let b be another finite real number. Then

$$[a, b] = (a, b) \cup \{a\} \cup \{b\}$$

is in $\mathcal{B}$ because it is a countable union of sets that we already know to be in $\mathcal{B}$. By similar reasoning, we can show that bounded, half-open intervals are in $\mathcal{B}$.

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(2.)

Statement

Every open set is an F_σ set.

Proof

Let $\mathcal{O}$ be an open set. Then we can write

$$\mathcal{O} = \bigcup_{k=1}^{\infty} (a_k, b_k),$$

where a_k, b_k are extended real numbers satisfying $a_k \leq b_k$. Note that we can write each (a_k, b_k) as

$$(a_k, b_k) = \bigcup_{n=1}^{\infty} \left[a_k + \frac{b_k - a_k}{3n}, b_k - \frac{b_k - a_k}{3n} \right].$$

So we can write

$$\mathcal{O} = \bigcup_{k=1}^{\infty} \left(\bigcup_{n=1}^{\infty} \left[a_k + \frac{b_k - a_k}{3n}, b_k - \frac{b_k - a_k}{3n} \right] \right).$$

A countable union of countable unions is still a countable union. Thus, $\mathcal{O}$ is a countable union of closed sets. Therefore, it is an F_σ set.

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(3.)

Statement

Let $A \subset \mathbb{R}$. Then for any $\varepsilon > 0$, there exists an open set $\mathcal{O}$ such that $A \subset \mathcal{O}$ and $m^*(\mathcal{O}) \leq m^*(A) + \varepsilon$.

Proof

Let $\varepsilon > 0$. First, suppose $m^*(A) = \infty$. Note that $\mathbb{R}$ is open and $A \subset \mathbb{R}$. We have

$$\begin{aligned} m^*(\mathbb{R}) &= \infty \\ &\leq \infty \\ &= \infty + \varepsilon \\ &= m^*(A) + \varepsilon, \end{aligned}$$

as required.

Next, suppose $m^*(A) < \infty$.

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(4.)

Statement

Suppose A, B are subsets of $\mathbb{R}$ and $m^*(A) = 0$. Then $m^*(A \cup B) = m^*(B)$.

Proof

Clearly $B \subseteq A \cup B$. Thus, by the monotonicity of outer measure, we have

$$m^*(B) \leq m^*(A \cup B).$$

We also have that

$$m^*(A \cup B) \leq m^*(A) + m^*(B) = 0 + m^*(B) = m^*(B),$$

by Proposition 3. Since $m^*(A \cup B) \leq m^*(B)$ and $m^*(B) \leq m^*(A \cup B)$, it must be the case that $m^*(A \cup B) = m^*(B)$.

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(5.)

Statement

Let A be the set of irrational numbers in $[0, 1]$. Then $m^*(A) = 1$.

Proof

Let B be the set of rational numbers in $[0, 1]$. Then we can write $[0, 1] = A \cup B$. Now, $\mathbb{Q}$ is countable, and since a subset of a countable set is countable, it follows that B is countable. Therefore, $m^*(B) = 0$.

Applying the result of Problem 4 (from the previous page), we see that

$$m^*([0, 1]) = m^*(A \cup B) = m^*(A).$$

But $m^*([0, 1]) = 1$. Thus, $m^*(A) = 1$.

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